

# Chapter 4:

## Hacking

### Using Hacking Tools

If you are proficient with and have hacking tools you can use an action to begin a hack, making a hacking tools check against a DC determined by the security of the device you are hacking. On a success you begin hacking that device. While hacking a device you cannot attempt to hack another device until the previous hack attempt ends.

### Uncontested Hacks

Most minor devices have little security to speak of and once in its system a hacker has full control over a device once a hack has begun. A hacker can continue hacking an uncontested hack indefinitely, and can perform any hacking maneuvers appropriate to that device without spending hacking countdown dice (see below).

**Locked Out.** On a failed hacking tools check for an uncontested hack, or when the hacker chooses to end the hack, the hacker is locked out of the device.

**Table: Uncontested Hacks**

| Device                    | Hacking DC |
|---------------------------|------------|
| Beverage vending machine  | DC 5       |
| Automatic door            | DC 10      |
| Communication device      | DC 10      |
| Computing device          | DC 12      |
| Private apartment         | DC 15      |
| Warehouse door            | DC 15      |
| Camera System             | DC 20      |
| Private encrypted datapad | DC 25      |

### Contested Hacks

More advanced devices are integrated with anti-hacking defenses or intelligent enough to actively resist intrusion. Once a device begins defending itself, it is only a matter of boldness and time before intruders are rooted out.

**Hacking Countdown.** After successfully making the initial hacking tools check to begin a contested hack, you create a countdown pool with a number of six-sided dice equal to the result of your hacking tools check divided by 5 (round down), added to your Intelligence modifier.

The dice within the dice pool are called hacking countdown dice. At the end of each of your turns that you spend hacking, roll these dice and remove any with a result of 6. Until a hacking countdown ends, the hacker can perform any hacking maneuvers appropriate to the hacked device.

**Locked Out.** On a failed hacking tools check, when there are no more hacking countdown dice, or when the hacker chooses to end the hack, the hacker is locked out of the device.

**Table: Contested Hacks**

| Device               | DC/Save  |
|----------------------|----------|
| Communication device | DC 10/+0 |
| Civilian vehicle     | DC 12/+1 |
| Computing device     | DC 12/+1 |
| Automated turret     | DC 15/+2 |
| Military vehicle     | DC 18/+3 |
| Hackable armor       | DC 20/+4 |
| Shielding Device     | DC 25/+6 |

## Hacking Lockouts

When a hacker is locked out of a device its internal security has identified them as an intruder, expelled the threat, and has gone on high alert.

A hacker that has been locked out of a device cannot attempt to hack that same device using the same hacking tools or method while it remains on high alert. In addition, a device that is on high alert gains an expertise die on saving throws made against hacking maneuvers.

The duration of a high alert varies between devices depending on their security sensitivity. The High Alert Duration table provides some examples but exactly how long a high alert lasts on any given device is up to the Narrator's discretion.

**Table: High Alert Duration**

| Device                       | High Alert Duration |
|------------------------------|---------------------|
| Private datapad              | 1d10 rounds         |
| Commercial hovercar          | 1d10 minutes        |
| Security cameras             | 1d10 hours          |
| Capital ship deck            | 24 hours            |
| Secure military installation | 1d10 days           |

## Networks and Terminal Access

Trying to penetrate security through a network can be infinitely more difficult than directly accessing a closed system through a terminal. By using hacking tools a hacker can attempt a hack on any device currently connected on the same network. Some networks are small and more closed off, consisting of only a single capital ship's systems or the systems of a single military installation, but others are vast

interconnected webs covering whole worlds or even solar systems.

**Network Access.** Checks made to initially access a device remotely using only a network are made with disadvantage. Once initial access is acquired, further checks are made normally.

**Terminal Access.** Checks made to initially access a device using a physical access point or directly connected terminal are made normally.

It is up to the Narrator how closed-off or easy to access a network is, but generally if a device is broadcasting or connected to any sort of internet or web it is accessible to anyone who can pick up a wireless signal (including anyone with hacking tools). Most ships have countless computer terminals and access points, while robots and other large machines typically only have a single access point. Often such access points are intentionally difficult to get to, either hidden behind panels or blocked by security measures. Most vehicles, starfighters, and constructs have their access points blocked in this way, and a thieves' tools check with a DC equal to their hacking DC is required before the device can be hacked directly.

## Hacking Capital Ships

Capital ships are great technological marvels containing numerous interconnected systems. For the purposes of hacking, each deck is considered to be a separate device. The hacking DC and saving throw bonus of a capital ship is shown in the ship's statistics.

## Counter Hacking

The same techniques used for hacking into a device can be used to lock someone out

of it. You can attempt to counter hacking as an action if:

- You are operating a device that is being hacked.
- You are operating in the same network as a device that is being hacked.
- You are the device being hacked.

When you attempt counter hacking, you make a computers check against the hacker's hacking maneuver DC. On a success you remove one of the countdown dice from that hack's dice pool. If you critically succeed, you remove 2 dice instead.

**Power Down.** You can also force a hack to end early by completely unpowering the hacked device. Devices must be completely severed from their power source to end a hack in this way as hackers can still access devices that are switched off but are still receiving power. Most small or handheld devices can be completely unpowered as an action, by pulling power cords or removing batteries. Larger devices such as starfighters, vehicles, and starship decks may be unpowered as an action but doing so requires access to the appropriate control panels and a successful DC 15 Engineering check. Construct creatures can voluntarily choose to unpower themselves as an action; if they do they are stunned until their power is restored. Devices can have their power restored as an action. A device is unusable while unpowered.

**Tracing.** Alternatively, you can attempt to trace the hacker rather than simply remove them from the system. On a successful check when counter hacking, you may choose to trace the hack instead of removing a countdown die. A critical success counts as 2 successful hacking traces instead of 1. If you successfully trace

the hack 3 times during the same hack, you learn the exact location of the hacker and what device they're using.

## Hacking Maneuvers

Access to a device is only half the battle—after getting into a system a hacker must figure out how exactly to play with their new toys. Hacking maneuvers are feats of technical prowess, clever tricks and methods for accessing information thought to be safe from systems said to be impenetrable.

A hacking maneuver usually requires a hacking tools check against the target device's hacking DC, or it allows the target device to make a Wisdom saving throw. For many targets, such as starships, the hacking DC and Wisdom saving throw bonus are already provided. For constructs such as androids or drones, the hacking DC is calculated as follows:

$$\text{Hacking DC} = 10 + \text{proficiency bonus} + \text{Wisdom modifier}$$

A hacking maneuver sometimes forces a target to make a saving throw to resist its effects. The saving throw DC is calculated as follows:

$$\text{Hacking Maneuver DC} = 8 + \text{your proficiency bonus} + \text{your Intelligence modifier}$$

Most hacking maneuvers also have a die cost representing the risks taken to achieve them. Whenever you perform a hacking maneuver during a contested hack, you must remove a number of countdown dice from the hacking countdown pool equal to the hacking maneuver's dice cost. If the hacking countdown does not have enough

remaining hacking dice in its dice pool to pay this cost, you cannot perform the hacking maneuver.

All hacking maneuvers may be performed by any hacker, but most can only be performed on the appropriate devices. The general types of devices appropriate for a hacking maneuver are listed in the hacking maneuver.

Hacking maneuvers are nonmagical and nonpsionic.

The target system's saving throw bonus is noted in Table: Contested Hacks. If the target of a device is held by another creature or operated by another creature (such as a pilot or helmsman), that creature makes any saving throws made to resist hacking maneuvers for the device.

Table: Hacking Maneuvers

| Maneuver           | Cost   |
|--------------------|--------|
| Blackout           | 2 dice |
| Bug the Line       | 1 die  |
| Copy Database      | 3 dice |
| Delete Data        | 1 die  |
| Digital Pickpocket | 2 dice |
| Digital Filter     | 4 dice |
| Disable Deck       | 3 dice |
| Drop Shields       | 3 dice |
| Evade Security     | —      |
| Freeze Engines     | 4 dice |
| Implant Data       | 3 dice |
| Jamm Comms         | 2 dice |
| New Target         | 1 die  |
| Open/Close         | 1 die  |
| Overclock          | 3 dice |
| Overheat           | 3 dice |
| Remote Control     | 4 dice |
| Scramble Sensors   | 2 dice |
| Search Files       | 1 die  |
| Self-Destruct      | 5 dice |
| Spam               | 1 die  |
| Spoof Readings     | 2 dice |
| Weapon Jam         | 2 dice |

## Encrypted Data

While some maneuvers provide access to data, if it is encrypted it will need to be decrypted.

Decrypting data, like deciphering codes or translating languages, is not a function of hacking. Once the encrypted information is extracted, an Investigation check is required, with the *deciphering* speciality being particularly relevant. The DC of that check will usually be the same as the hacking DC of the device, unless the Narrator rules otherwise.

### Blackout (2 dice)

*Action, save (camera systems, construct creatures)*

Crashing a visual feed is one of the oldest tricks in a hacker's book. The device makes a Wisdom saving throw or is blinded and cannot transmit visual information so long as you are hacking the device.

### Bug the Line (1 die)

*Action, check (any ship deck, communications devices)*

You subtly patch yourself into lines of communication thought to be secure. So long as you are hacking the device you are able to see and hear anything transmitted through it. If you hack a ship's deck in this way, you are instead able to see and hear everything transmitted through that ship's comms (including correspondence between it and any starfighters launched from it).

### Copy Database (3 dice)

*Action, check (communications devices, computing devices)*

Passwords, dates, names, bank accounts—all sorts of data is yours for the taking once you've cracked a system. You completely copy and download the data contained within the device into your hacking tools. If the Narrator determines that the

### Delete Data (1 die)

*Action, check (any ship deck, camera systems, communications devices, computing devices)*

With professional subterfuge sometimes you just need to wipe the records clean. You remove data within a device, such as wiping recent camera footage or erasing the contents of a ship's manifest. The result of your hacking check is also used as the DC of Investigation checks made to determine that the information was deleted by a hacker and isn't simply missing or corrupted.

### Digital Pickpocket (2 dice)

*Action, check (communications devices, computing devices)*

Most accounts lock up tight when an attempt is made to empty them out so the smartest thieves collect their ill-gotten gains in little innocuous bits. Transfer up to 5% of the credits in an account linked to the hacked device into an account of your own. Software on a monied account is usually very tight however, and when you begin this hack your hacking countdown becomes a fast countdown (dice are removed from the pool on a result of 4, 5, or 6).

### Digital Filter (4 dice)

*Action, save (camera systems, construct creatures)*

You can use high speed editing viruses to literally strip content out of mechanical eyes. The device makes a Wisdom saving throw or you make any number of creatures and objects unseen to it. The device treats your selected creatures and objects as invisible, and cannot visually perceive them so long as you are hacking the device. Any visual information the device transmits appears as normal, except your chosen creatures and objects are unseen so do not appear within them.

### Disable Deck (3 dice)

*Action, check (any ship deck)*

Most ship systems have internal lockouts to prevent hacking—it'd be a shame to trigger that intentionally to lockout the intended users. No actions or ship maneuvers can be made using the device while you are hacking it. Once your hack ends, the device regains full functionality.

### Drop Shields (3 dice)

*Action, check (construct creatures, forcefields, shielding devices, shield generation decks, starfighters)*

Shields and forcefields are excellent defenses in the physical world but of little use in the digital realms. Any temporary hit points the target has gained from a shield or forcefield are immediately lost. Devices hacked in this way cannot gain temporary hit points from their shields or forcefields while you are hacking the device. If the hacked device is forcefield armor, the AC provided by the hacked armor is reduced to 10 + the wearer's Dexterity modifier while it

is being hacked. Once your hack ends, the forcefield regains full functionality.

### Evade Security (0 dice)

*Action, check (any device)*

Sometimes you just need a little more time with system access and that means evading digital security measures. You add one additional hacking countdown die to the hacking countdown. If you critically succeed on this check, instead add 1d4 hacking countdown dice to the hacking countdown.

### Freeze Engines (4 dice)

*Action, check (engine decks, starfighters, vehicles)*

Engines are so complex that all it takes is bypassing the internal safety protocols to make something go wrong. The device's engines cease to function while you are hacking it. Devices hacked in this way cannot move, perform any starfighter maneuvers, or make any system maneuvers that require movement. If the hacked device is currently within a planet's atmosphere or gravity well it begins falling downward at a rate of 500 feet per round. Once your hack ends, the engine regains full functionality.

### Implant Data (3 dice)

*Action, check (any ship deck, communications devices, computing devices)*

Putting data where it shouldn't be without leaving a digital fingerprint is a tricky task, but when you cover your tracks most people simply believe what's on their screens. You implant data within a device, such as adding names to a company roster or falsifying records. The result of your hacking tools check is also used as the DC of

Investigation checks made to determine that the information you implanted is inaccurate and has been tampered with.

### Jam Comms (2 dice)

*Action, save (any deck, communications systems, construct creatures)*

Information and coordination are key to any battle—turn those off and it's chaos. The device or its operator makes a Wisdom saving throw. On a failed save, while you are hacking it the device is deafened, it cannot transmit audio information, and any signals it's connected to are jammed and cannot send or receive data transmissions. If you hack a capital ship deck in this way this effect applies to all of those ship's comms and ship comms held by creatures crewing it. Once your hack ends, the communications system regains full functionality.

### New Target (1 die)

*Bonus action, check (construct creatures, starfighters, vehicles, weapon decks)*

In the chaos of battle it can be easy to miss a few subtle tweaks to the targeting system. The next time the device makes an attack, it or its operator makes an Intelligence saving throw or you select a new target for that attack. New targets must be eligible within reach or range of the triggering attack.

### Open/Close (1 die)

*Bonus action, check (doors, vehicles)*

Sometimes you just need access or to deny access to others. You force the door or its doors to either unlock and open or to close and lock shut.

### Overclock (3 dice)

*Action, check (construct creatures, hackable armors)*

Most machine intelligences have untapped processing potential just waiting to be unlocked at the mere cost of system integrity and operating life. While you are hacking the device it gains an expertise die on attack rolls, ability checks, and saving throws. However, for the duration of the hack the device takes 2d4 ongoing lightning damage plus an additional 1d4 lightning damage for each round past the first. Once your hack ends, the device returns to normal functionality.

### Overheat (3 dice)

*Action, check (computer systems, construct creatures, hackable armors, starfighters, vehicles)*

By disabling enough safeguards you can cause the inner workings of most devices to grind and burn. The device takes 2d6 ongoing fire damage while you are hacking it.

### Remote Control (4 dice)

*Action, check (construct creatures, hackable armors, starfighters, vehicles)*

Most controls don't have mechanical meaning anymore, it's all just electrical signals sent to moving parts—why not send those signals from somewhere else? You bypass an internal safeguard. Once you successfully bypass 3 internal safeguards you take control of the device. When you take control of a starfighter or vehicle in this way you are treated as if you were the operator while you are hacking it. You can direct the actions of a construct creature hacked in this way as a bonus action while

you are hacking it. Once your hack ends, the device returns to normal functionality. Construct creatures are aware that they have been hacked in this manner.

### Scramble Sensors (2 dice)

*Action, save (construct creatures, hackable armors)*

You fill the device's sensors with so much noise that it temporarily loses grip on its Subroutines. The device makes a Wisdom saving throw or is confused while you are hacking it. Once your hack ends, the device returns to normal functionality.

### Search Files (1 die)

*Bonus action, check (communications devices, computing devices)*

All sorts of valuable and personal data is yours for the taking once you've cracked a System. You look through the device's data as if you had direct access to it. You retrieve one specific piece of information.

### Self-Destruct (5 dice)

*Action, check (bridge decks, engine decks)*

Most capital ships either have self-destruct emergency systems by design or react catastrophically when the right combination of failsafes go awry. You bypass an internal safeguard. Once you successfully bypass 3 internal safeguards you activate the device's self-destruct sequence. You may choose the duration of this self-destruct sequence (minimum 30 seconds and maximum 10 minutes) after which the capital ship explodes.

### Spam (1 die)

*Bonus action, check (communications devices, computing devices, construct creatures, hackable armors starfighters)*

You fill the device with pop-up images and nonsense, drowning out its actual functions with useless and distracting chatter. While you are hacking the device it is rattled, or if the device is operated the creature operating it is rattled until it is no longer operating the device. Once your hack ends, the device returns to normal functionality.

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## Spoof Readings (2 dice)

*Action, check (science decks, scientific devices)*

Armies sometimes march on one piece of trusted data and scientific equipment is just so precise. All it takes is a little unseen tweak to make a huge difference. You determine the result of the next test or Scan the device makes. Your fictitious result must be a possible result of this next action, but you can determine the displayed result. Examples include detecting dangerous radiation where there is none, concealing implants from an x-ray scan, or fabricating different results from a ship Scan.

## Weapon Jam (2 dice)

*Action, save (construct creatures, starfighters, vehicles, weapon decks)*

Even the most advanced weaponry is made up of multiple moving parts with their own exploitable vulnerabilities. The next time the device makes an attack or an attack is made using one of the device's weapons, it or its operator makes a Dexterity saving throw. On a failed save the weapon jams and cannot be used to make attacks while you are hacking the device. The device or the device's operator may then immediately make an attack using a different weapon with the same Attack action, assuming they have another weapon to use. Once your